

Automated Target Measurement

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Our Client

Analog Devices is a semiconductor designer and manufacturer based in the United States, with a product line that includes various ICs including analog-to-digital/digital-to-analog converters, power management products, digital signal processors, and amplifiers.

Background

In Analog's wafer manufacturing process, they use disks of pure metal (targets) as source material to deposit that metal onto a silicon wafer in a process called physical vapor deposition (PVD). As a part of PVD, it is critical that a target does not become too thin and develop holes ("burn through") or expose the wafer to non-pure metal on the backside of the target.

To estimate a target's lifespan, Analog uses a manual measurement process that is time-intensive and requires extensive training to be done in a repeatable way. This limited the number of targets Analog was able to measure to calculate their estimate of target life, and forced them to use a more conservative estimate of target lifespan to prevent burn-throughs, as they had on a small sample of data to work off.

Project Goal

Analog's goal for our project was to fully automate the target measurement process, maintain the same $1\mu\text{m}$ precision, and reduce the complexity of the procedure to measure a target. In doing so, this would enable Analog to measure more targets, enabling them to make a more precise estimate of target lifespan.

Targets are expensive, with pricing ranging from thousands to hundreds of thousands of dollars, so using each target to its maximum lifespan without destroying wafers is of high importance.

Original System

Analog's existing system for measuring targets included a target jig with holes for mounting a Keyence laser track at various cross-section angles of the target.

As the target jig was not perfectly level with the laser track, shims were manually placed under the laser track. This leveling step was the most time consuming part of the original procedure, and needed to be performed for every cross section measurement.

3 distinct pieces of software were used to perform the measurement: the laser control software, the motor control software (Windows 8 only) to move the laser, and Excel to process the data.



Fig. 1: Pre-existing ADI measurement setup

Prototype Designs

The first prototype centered around re-using the measurement jig that was given to us by Analog. The static bottom plate would be removed and replaced by a motorized turntable. This proved inefficient since the laser would still need to be removed for each new target, as there was no way to insert a target without removing the laser gantry.

The second prototype, which shares many similarities with the final version, is a "pizza-oven" style device with a turntable in the center. A target is slid onto the turntable, is aligned by guides on the turntable face, and then measured by the laser which tracks across on a fixed ball screw drive. This had many advantages over the first prototype including ease of target insertion/removal, protection of the delicate laser, and ease of access to all mechanical and electrical components.

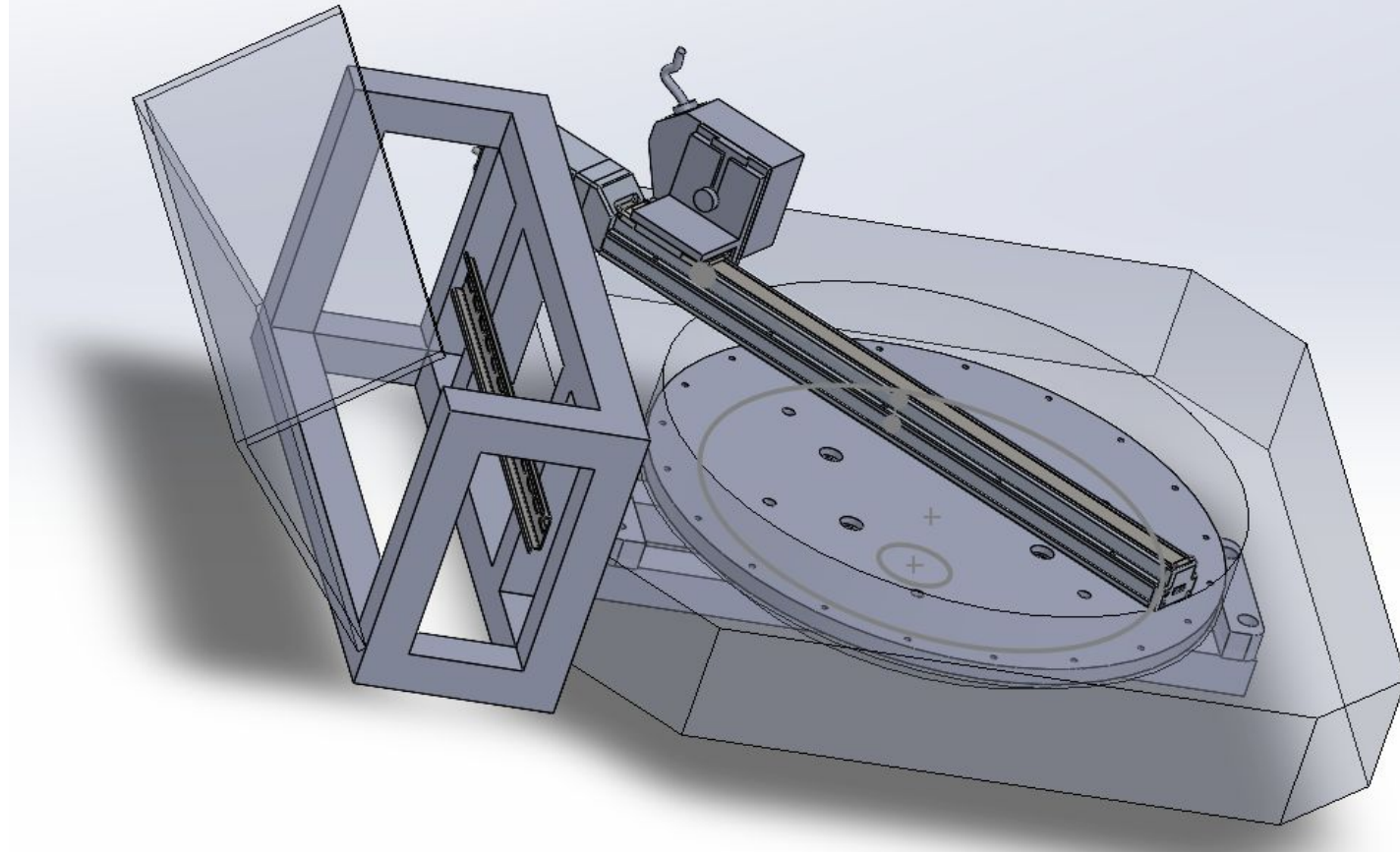


Fig. 2: Rendering of potential modifications to existing jig

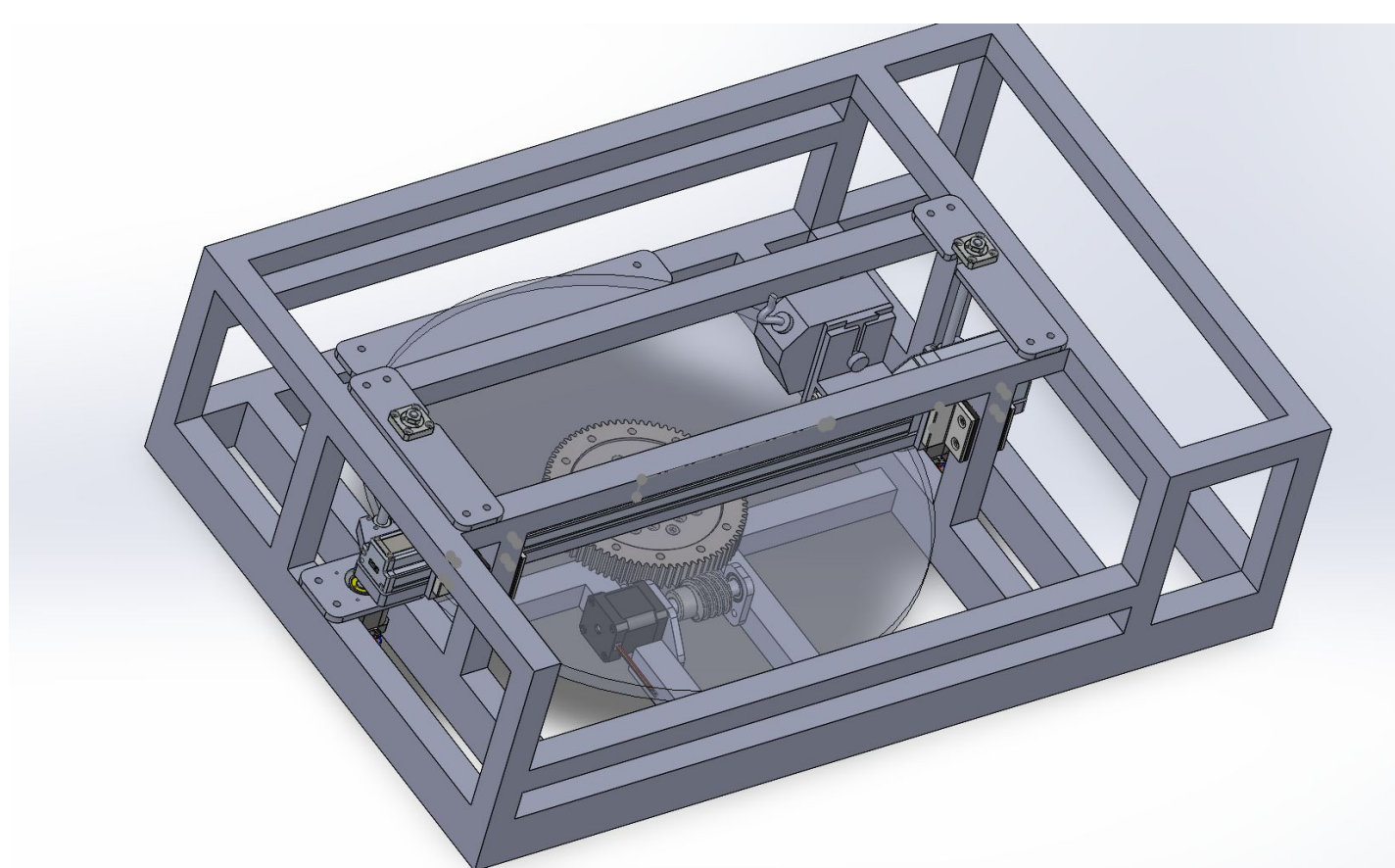


Fig. 3: "Pizza oven" prototype with turntable bearing

Final Design & Results

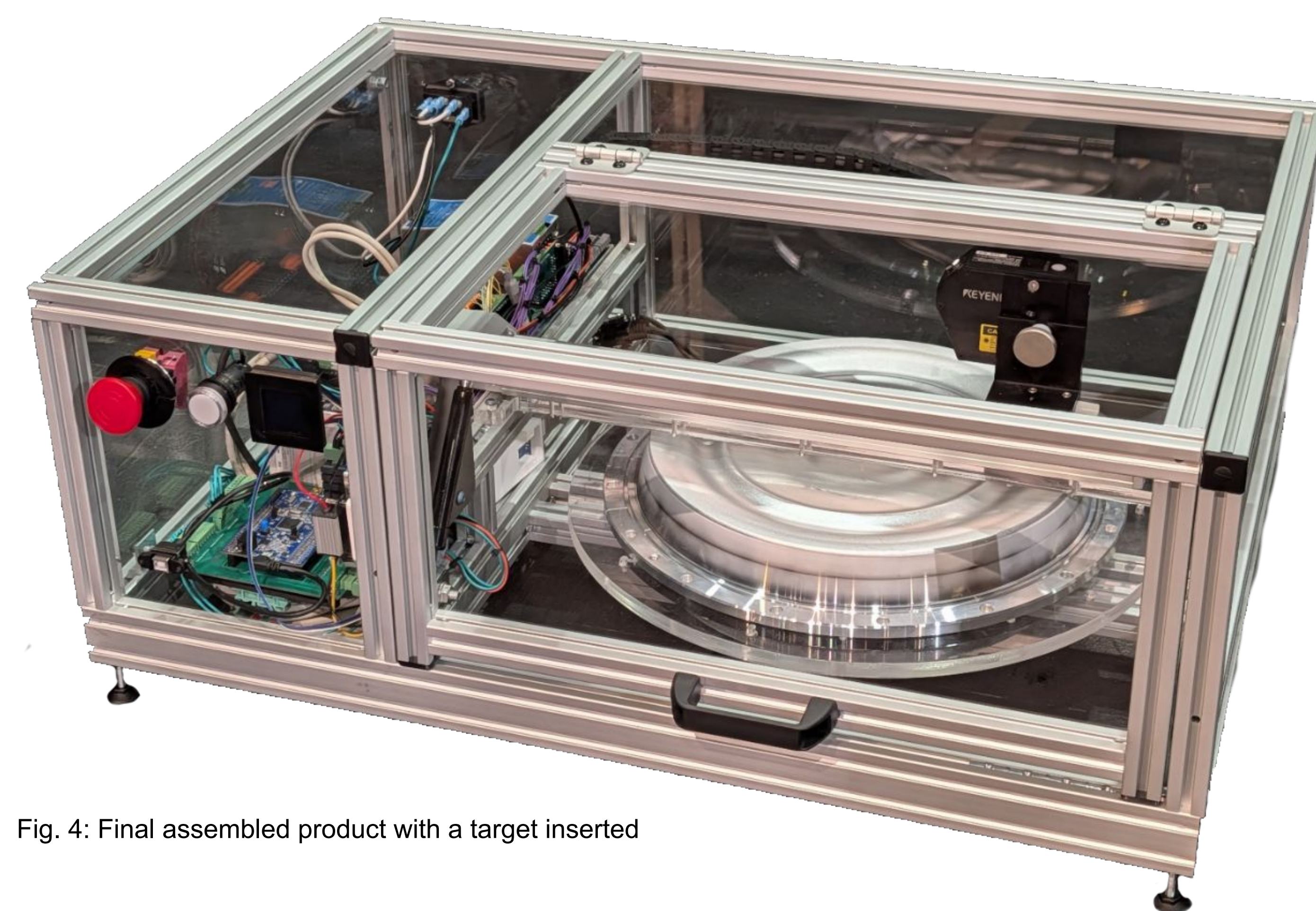


Fig. 4: Final assembled product with a target inserted

The final design features an 8020 aluminum extrusion frame with a rotating turntable. The design is segmented into two sections: the measurement section, which houses the laser gantry and rotating turntable for the target, and the control section which houses all the electronics and control systems for the gantry and rotation motors.

An STM32F4 microcontroller provides a USB interface that can be accessed by our custom-designed software on any modern Windows computer. This software can control every part of the device, including the laser, motors, and the user interface on the front of the device.

The software features two modes: an automatic mode, which executes a predefined procedure from a user-editable config file, and a manual mode, which allows the user to individually control any part of the device for debugging.

Once a user takes a measurement, the data is sent over USB to the computer, and then analyzed using several Python scripts to automatically level and normalize the measurements, and then display the results to the operator.

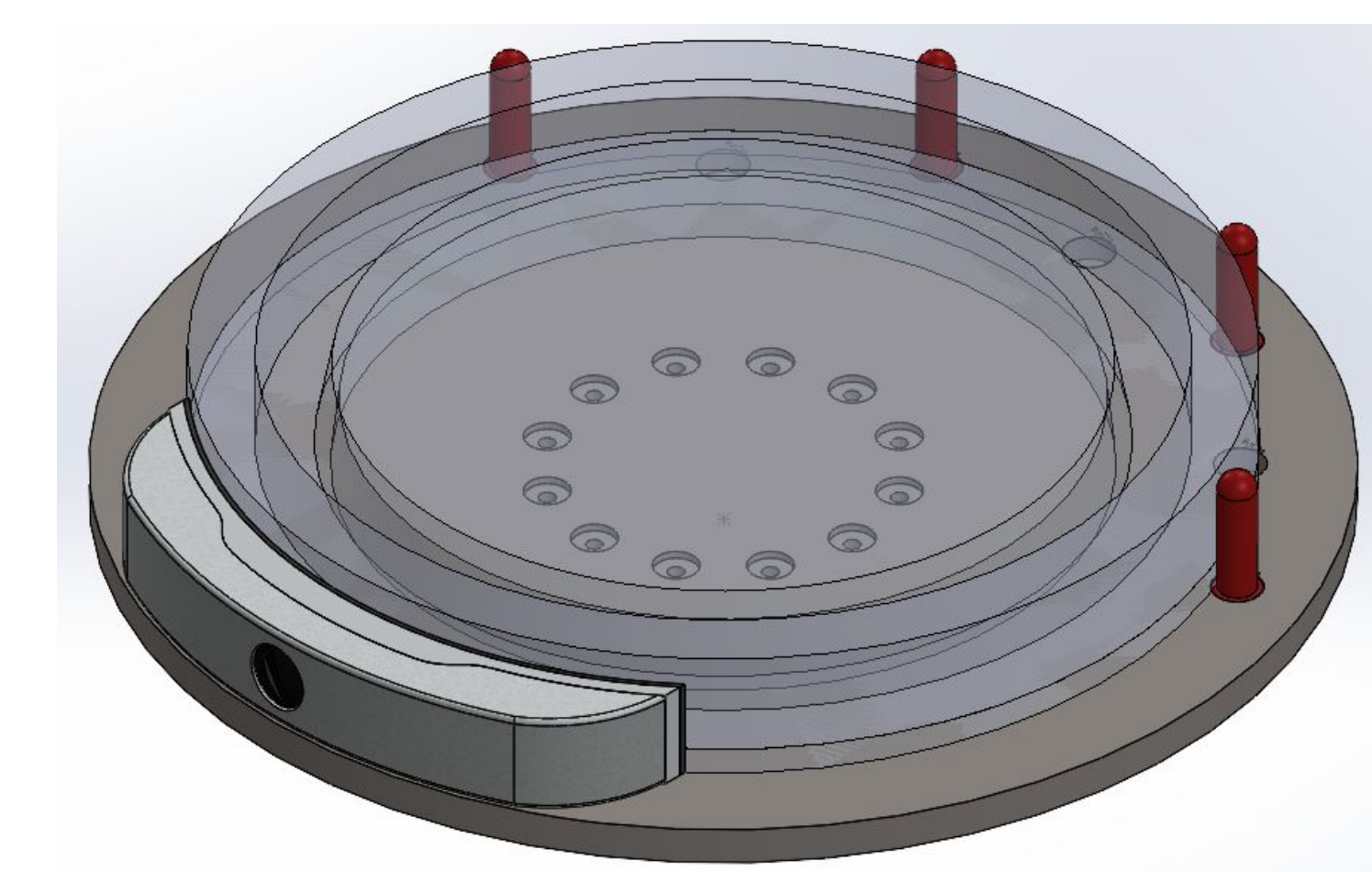


Fig. 5: Final target clamp mechanism

Leveling the Target

To mechanically level each target proved to be feasible, but also complex and expensive. We decided to research whether software could account for an out-of-level target and return the same result as a leveled target, and were able to create a reliable algorithm.

When a target is inserted into the device, we get the depth measurements from the laser, and then look for the measurement ring on either side of the target. This ring is the reference which is used to level the target manually and also through software. A first derivative test is used to find the edges of the ring, and then a point 2mm from the inner edge of the ring is chosen on both sides. These two points form a line, which is used to adjust the rest of the data points in between, thus leveling the disk.

The example below shows a manually leveled vs. an extremely out-of-level target measurement, and the output of the software leveling algorithm.

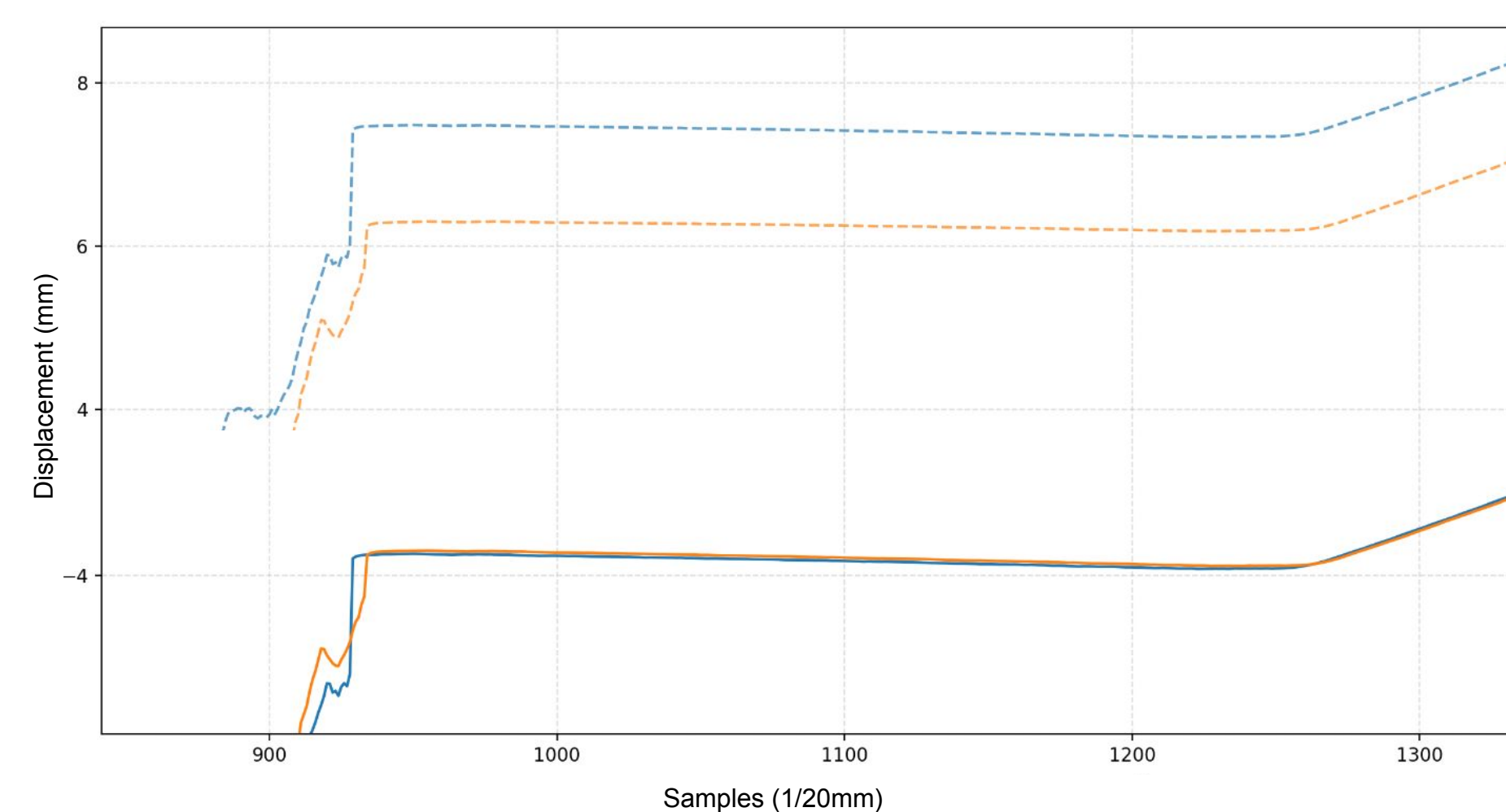


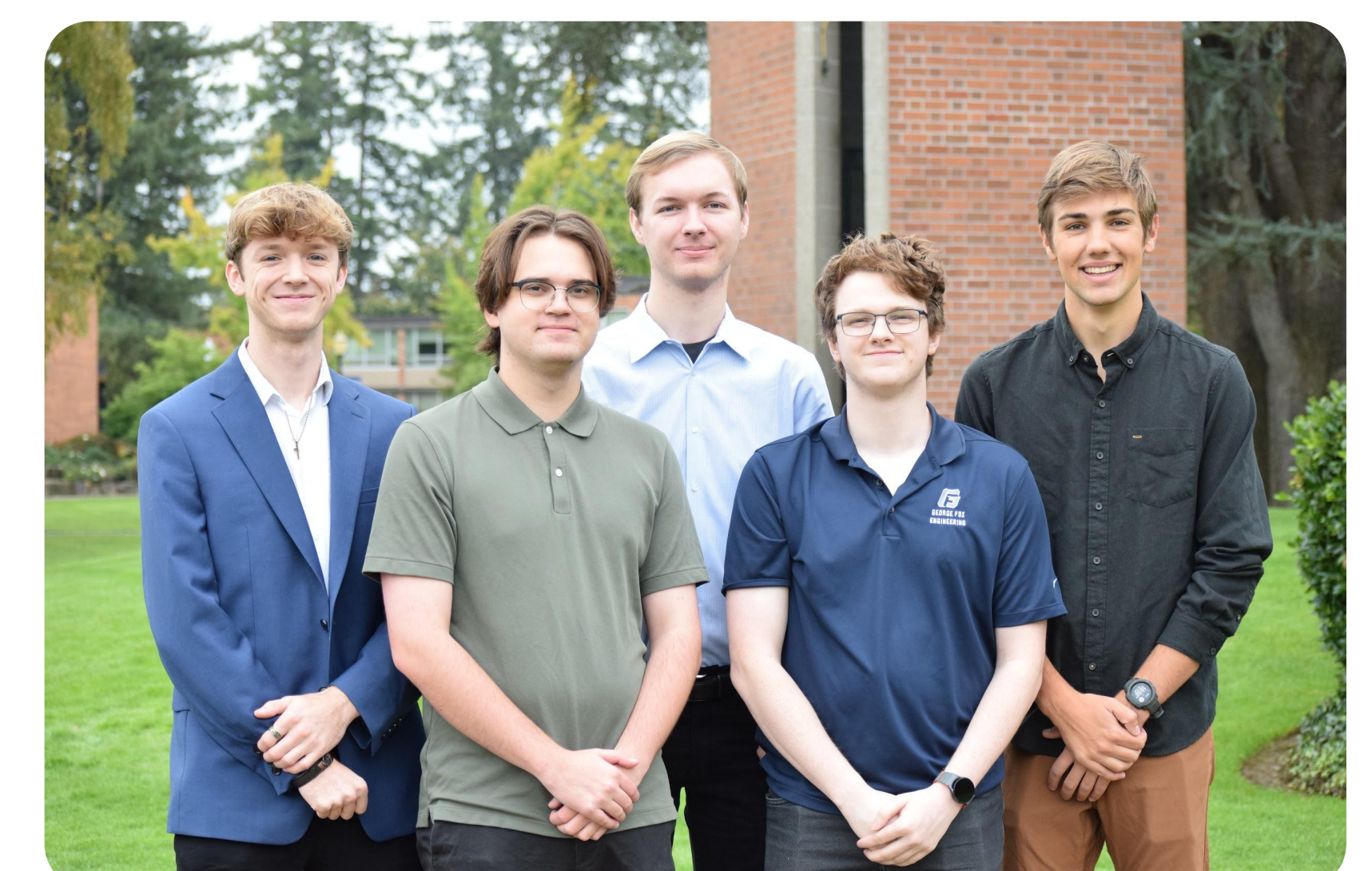
Fig. 6: Before & after the leveling of two target measurements

Future Work

Once data is collected by the target measurement device, the next step is to create a model of target machine hours to target wear/remaining material, and compare Analog's existing target lifespan estimation to the results from more frequent measurements to see if the target lifespan can be increased.

To improve the user experience of the device, a drawer-style mechanism to assist in loading/unloading the target would also be beneficial, as the targets can be quite heavy.

Team



Team Members (from left to right):
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